SPS-VARIMF User Manual

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1. Introduction

The **Stellar Population Synthesis for Varying Initial Mass Function (SPS-VARIMF)** model is designed to compute the spectral and photometric evolution of stellar populations across a broad parameter space. This model is explicitly developed to investigate the impact of a non-universal Initial Mass Function **(IMF)** on key galactic properties, including integrated luminosities, stellar masses, and the fractional contributions of stellar remnants. By incorporating IMF variations, SPS-VARIMF provides a more comprehensive framework for studying the evolutionary pathways of galaxies and their observable characteristics.

The model specifically focuses on galaxies in which the IMF evolves as a function of star formation history (SFH) and metallicity, offering a more realistic representation of stellar population synthesis. It produces outputs that include high-resolution spectra and magnitudes in multiple photometric bands, enabling detailed comparisons with observational data.

IMF Variation in SPS-VARIMF

In SPS-VARIMF, the **galaxy-wide IMF (gwIMF)** is allowed to vary at each time step based on the galaxy's star formation rate (SFR) and the gas-phase metallicity at that epoch. The code includes both an **invariant IMF** and the **Integrated Galaxy-wide IMF (IGIMF)** framework, but it can also accommodate any other variable IMF prescription through private communication. This flexibility makes **SPS-VARIMF** a powerful tool for exploring the consequences of IMF variability in different astrophysical contexts.

Metallicity Evolution

The model simulates metal enrichment processes under the assumption of a closed-box system, meaning no gas inflow or outflow is considered. The metallicity evolution is inherently dependent on the adopted **star formation efficiency (SFE)**, which governs the rate of metal production and retention. However, users have the flexibility to assume a constant metallicity if desired, allowing for controlled comparisons and alternative modeling approaches.

Dust Attenuation

An optional dust attenuation component is included in the model, implemented using a **power-law formalism** to account for the wavelength-dependent absorption and scattering of light by interstellar dust. This feature allows users to incorporate the effects of dust on spectral energy distributions (SEDs) and photometric properties, providing a more realistic comparison with observed galaxy spectra.

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2. Setting Up the Model

Before compiling, ensure that the SPS-VARIMF directory is located in the same parent directory as the following required folders:

- **SPECTRA**
- **ISOCHRONS**
- **FILTERS**
- **YIELDS**
- **INDICES**
- **OUTPUTS**

2.1 Compiling the Code

The code includes a **Makefile** to facilitate compilation. Before proceeding, ensure that your system has a Fortran compiler installed (e.g., gfortran). You may need to modify the Makefile to specify the correct Fortran compiler for your system

1. **Clean any previous builds** :

   make clean

2. **Compile the program**:

   make

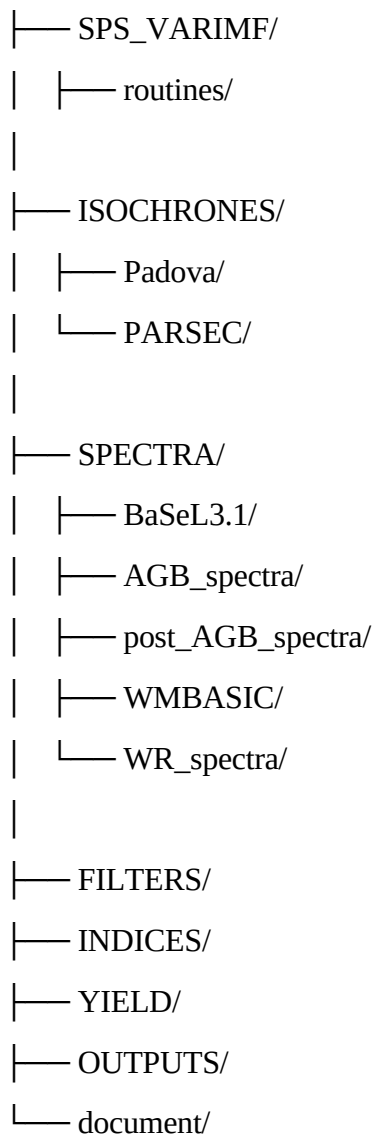
This will generate an executable file named **main.exe** in the SPS-VARIMF directory.

3. **Running the executable:**

   ./main.exe

2.2 Directory Structure

The program relies on a specific directory layout for its input files:



2.3 Adjust parameters :

To properly configure the SPS-VARIMF model, key simulation parameters must first be defined and initialized in **params.f90.**

Inorder to adjust optional cases

Dust attenuation (dust_atten):

= 0 : Disabled
= 1 : Enabled (power-law attenuation)

Remnant mass calculation (remnant_cal):

= 0 : Renzini et al. (1993) prescription
= 1 : Spera et al. (2015) prescription

Metallicity evolution mode (Z_MODE):

= 0 : Constant metallicity
= 1 : Metallicity computed at each step from gas enrichment

OB stars count calculation (OB_stars):

= 0 : Do not compute the number of OB stars
= 1 : Compute the number of OB stars

Compute indices (index_cal):

=0 : Do not compute indices
=1 : compute indices

Stellar yields (YIELD_TYPE):

= 1 : Portinari et al. (1998)
= 2 : Marigo (2001)
= 3 : Marigo (2001) for M < 8.5 Msun
& Portinari et al. (1998) for M > 8.5 Msun

Type Ia SN yields (yield_reference_SNIa):

= 1 : Seitenzahl et al. (2013)
= 2 : Iwamoto et al. (1999)
= 3 : Iwamoto et al. (1999) W70 model
= 4 : Iwamoto et al. (1999) W7 model
= 5 : Iwamoto et al. (1999) WDD3 model

2.4 Model Configuration Parameters:

Modify the simulation parameters in the **main.f90** program to setup the model. The key parameters include:

Initial Mass Function (IMF) (`imf_type`)

Specifies the type of IMF used in the simulation:

- `2` → **Canonical IMF (Kroupa 2001)** – A standard IMF with fixed power-law slopes.
- `3` → **Integrated Galactic IMF (IGIMF)** – An SFR- and metallicity-dependent IMF.
- `5` → **Top-Heavy IMF** – A flatter IMF favoring high-mass star formation.

Isochrones (`isoc_type`)

Determines the stellar evolution tracks used:

- `1` → **Padova (2007)**
- `2` → **PARSEC (2022)**

Spectral Library (`spec_type`)

Specifies the spectral database used for stellar populations:

- `1` → **BaSeL spectral library** (default)
- `2` → **Miles spectral library** *(in progress, not yet available)*

Star Formation History (SFH) (`sfh_type`)

Defines how the galaxy forms stars over time:

- `0` → **Single Stellar Population (SSP)** – All stars form in a single burst.
- `1` → **Constant Star Formation Rate (SFR)** – Stars form at a steady rate over time.
- `2` → **Exponentially Declining SFR** – Star formation rate decreases exponentially.
- 3 → **Eappen et al. 2022 SFR** – time-dependent evolution of the SFR according to the Eappen et al. (2022) prescription..
- `4` → **Delayed-Tau SFH** – A rising SFR phase followed by exponential decline.

Additional SFH Parameters:

- **e-folding timescale (`tau`)** (for `sfh_type = 2` and `4`):
  - Range: `0.001 – 1000.0` Gyr
- **The time of the peak of SFH (`tpek`)** (for `sfh_type = 3`):
  - Defined in **Gyr**, based on the age of the universe.
- **The star formation timescale (`dTm`)** (for `sfh_t = 3):`
  - Defined in **Gyr**.
- **Star formation start and truncation time (`Tstart`, `Ttrunc`)**:
  - Default: `Tstart = 0.0`, `Ttrunc = 0.0`
  - Defined in **Gyr**, based on the age of the universe.

Metallicity (`zmet_ini`)

Specifies the initial metallicity index:

- Choose a value from **1–22**, corresponding to metallicity values in **Table 1**.
- `(Z_MODE=0 )` **Constant metallicity** throughout the simulation (`zmet = zmet_ini`).
- `(Z_MODE=1 )` **Self-enrichment**: Metallicity evolves over time due to stellar feedback.

Galaxy Mass (`M_galaxy` or `M_UCD`):

Defines the total stellar mass formed over time:

- `M_galaxy` → Total stellar mass of a galaxy (in solar masses).
- `M_UCD` → Monolithic formation of **ultra-compact dwarf galaxies (UCDs)**, following a top-heavy IMF.

Star Formation Efficiency (f_star):

Determines how efficiently gas converts into stars:

$$f_{star} = M_{galaxy} / M_{gas}$$

- Valid range: **`0 < f_star < 1`**

3. Initial Mass Function (IMF) and galaxy-wide IMF (gwIMF)

3.1 The Initial Mass Function (IMF)

The **Initial Mass Function (IMF)** describes the mass distribution of stars formed in a single star formation event within a region approximately **1 parsec** in size. As a fundamental property of stellar populations, the IMF determines the relative abundance of stars of different masses, directly influencing the rate of supernova explosions and subsequent stellar feedback, chemical enrichment processes, and the overall luminosity of stellar populations and unresolved objects.

As a result, the IMF plays a central role in many astrophysical fields, including galaxy evolution, stellar dynamics, and cosmology. For simple stellar populations (SSPs), the IMF can be chosen to be either an invariant canonical IMF or a metallicity- and density-dependent top-heavy IMF. The canonical IMF represents a universal, fixed mass distribution, while the top-heavy IMF accounts for environmental dependencies, leading to an increased fraction of high-mass stars in metal-poor and high-density regions.

This flexibility allows the model to be applied to **monotonically forming stellar populations**, such as star clusters, where all stars are assumed to form in a single burst. Additionally, a **top-heavy** IMF may be appropriate for specific objects such as ultra-compact dwarf galaxies (UCDs), where observational and theoretical studies suggest that variations in the IMF may influence their mass-to-light ratios and stellar populations.

3.2 Galaxy-Wide Initial Mass Function (gwIMF)

The **Galaxy-Wide Initial Mass Function (gwIMF)** is the integrated IMF across an entire galaxy, rather than an IMF specific to an individual star-forming region. It defines the total stellar mass distribution, the formation rates of massive stars, supernova occurrence, and the photometric and chemical properties of galaxies.

In its simplest form, the gwIMF is often assumed to be universal, meaning it follows a single, invariant functional form across all galaxies, regardless of their star formation history (SFH) or metallicity. This assumption has been widely used in stellar population synthesis models and cosmological simulations. However, observational and theoretical evidence suggests that the IMF may vary systematically with environmental conditions such as **star formation rate (SFR)** and **gas-phase metallicity**.

One of the most well-established non-universal IMF frameworks is the **Integrated Galaxy-wide Initial Mass Function (IGIMF) theory (** Weidner et al. 2013; Kroupa et al. 2013; Yerabkova et al. 2018, Zonoozi et al. 2018, Yan et al. 2021, Zonoozi et al. 2025). In this approach, it is assumed that all stars form in embedded clusters (Kroupa & Weidner 2003; Dinnbier et al. 2022)**,** making the gwIMF an integral over all embedded clusters formed at a given time.

In this model, the IMF in individual star clusters depends on local conditions such as SFR and metallicity, leading to a systematically varying gwIMF across different galaxies. This variation impacts mass-to-light ratios, supernova feedback, and chemical enrichment, making the IGIMF approach more realistic for modeling galaxy evolution. This self-regulated star formation process, as implemented in SPS-VARIMF, ensures that the gwIMF is dynamically responsive to galactic conditions, providing a more realistic representation of stellar populations in galaxies.

3.3 The gwIMF in SPS-VARIMF

The SPS-VARIMF model allows users to explore both **universal** and **non-universal** (IGIMF-based or other variable IMF formulations) to investigate their impact on galactic properties. This implementation allows for a more realistic representation of stellar populations, significantly impacting predictions of stellar feedback, supernova rates, metal enrichment, and photometric evolution in galaxies.

The program supports:

- **Canonical IMF**: A Kroupa-like IMF with fixed slopes (alpha1 for low-mass stars, alpha2 for high-mass stars).
- **IGIMF**: An IMF that varies with SFR and metallicity.
- **Top-Heavy IMF**: A flatter IMF for extreme environments.

Subroutines such as IMF_NORM, IGIMF_NORM, and IMF_TOPHEAVY handle IMF normalization and computation of stellar remnants.

The GWIMF code[1] is used to compute the galaxy-wide initial stellar mass functions based on IGIMF theory for different adopted star formation rate or metallicity.

Implications of IGIMF:

- **Low SFR galaxies** have steeper gwIMFs (i.e., they form fewer high-mass stars), leading to reduced supernova rates and weaker chemical enrichment.
- **High SFR galaxies** have a flatter gwIMF, allowing for enhanced formation of massive stars and higher chemical enrichment rates.
- The IGIMF naturally explains observed trends in galaxy evolution, such as the **mass-metallicity relation** and variations in the Hα-to-UV ratio in star-forming galaxies.

This framework is implemented in the SPS-VARIMF model, allowing users to explore how IMF variations affect galaxy properties, such as stellar populations, chemical evolution, and feedback processes and how underlying IMF influence key galactic observables and compare them to empirical data.

1 https://github.com/ahzonoozi/GWIMF

4. Stellar Evolution Tracks and Isochrones

We provide an option to use **isochrones** based on **Padova** or **PARSEC** models[2]. These theoretical models describe stellar evolution in terms of luminosity, temperature, and internal structure, making them essential for stellar population studies, galactic evolution research, and star clusters.

Covering a wide range of stellar masses (0.1 $M_\odot$ to 350 $M_\odot$) and metallicities, they enable simulations of stellar populations across different galactic environments, from metal-poor early-universe stars to younger, metal-rich clusters. These models incorporate updated physics, including mass loss, convection, and opacity adjustments, ensuring high accuracy for astrophysical applications.

This code allows users to include isochrones from **Padova** or **PARSEC** models. These isochrones are computed for ages ranging from about 1 Myr to 14 Gyr and across various metallicities from Z=0.0001 ([Fe/H]=-2.2) up to Z=0.06 ([Fe/H]=+0.5). The corresponding metallicity indices are listed in Table 1, facilitating the modeling of stellar populations in different galactic environments.

5. Spectral Libraries

The program uses the **BaSeL** spectral library as a **primary source** for general stellar spectra. However, for **special types of stars**, spectral data are obtained from specific references.

1. **BaSeL:** General stellar spectra.
2. **TP-AGB (Lancon & Mouhcine 2002):** Spectra for O-rich and C-rich Asymptotic Giant Branch (AGB) stars.
3. **Post-AGB (Rauch 2003):** NLTE spectral models for post-AGB stars.
4. **Wolf-Rayet Stars (Smith et al. 2002):** Spectra for Wolf-Rayet (WR) stars.
5. **OB stars (Eldridge et al. 2017):** High-resolution spectral models for OB stars.

Subroutines such as SPEC_AGB.f90, SPEC_PAGB.f90, and SPEC_WR.f90 process and interpolate these libraries onto the wavelength grid.

6. Star Formation History (SFH)

The SFH defines how the galaxy forms stars over time. Supported types include:

- **Constant SFR**: Star formation proceeds at a constant rate.
- **Exponential Decline**: SFR declines exponentially with time.
- **Delay-Tau Model**: A combination of a rising phase followed by exponential decay.

The SFH subroutine (sfh_gal.f90) calculates the mass fraction and star formation rate over specified time intervals.

2 http://stev.oapd.inaf.it/cgi-bin/cmd

7. Metallicity Evolution

The current implementation uses a **closed-box model** for metallicity enrichment, where all gas remains within the galaxy. The enrichment process is driven by stellar yields, read from the YIELD/ folder. Future updates will incorporate **gas inflow models.**

8. Chemical Evolution and Stellar Yields

The chemical evolution module follows the enrichment of the interstellar medium by successive generations of stars. The current implementation adopts a closed-box model, where the total gas mass remains within the galaxy and no gas inflow or outflow is considered.

Metal enrichment is computed from stellar ejecta using different stellar yield prescriptions available in the code. The yields define the amount of chemical elements released by stars of different masses and by Type Ia supernovae during their evolution.

The chemical evolution module tracks the enrichment of several chemical elements produced by stellar evolution and supernova explosions. The mass of the most abundant elements H, He, C, N, O, Ne,Mg, Si, S, Ca, Fe, are traced separately.

Supported stellar yield models include:
• Portinari et al. (1998): Massive-star nucleosynthetic yields.
• Marigo (2001): Low- and intermediate-mass stellar yields.
• Hybrid Model: Marigo (2001) yields for stars with M < 8.5 Msun combined with Portinari et al. (1998) yields for M > 8.5 Msun.

Type Ia supernova enrichment can be computed using several yield prescriptions:
• Seitenzahl et al. (2013)
• Iwamoto et al. (1999) standard model
• Iwamoto et al. (1999) W70 model
• Iwamoto et al. (1999) W7 model
• Iwamoto et al. (1999) WDD3 model

The stellar yield tables are stored in the YIELD/ folder

9. Filters and Photometry

The model computes magnitudes in 143 photometric bands, whose filter profiles are stored in the FILTERS/ folder and the filter order displayed in the FILTER_LIST.dat file. The Filtr.f90 subroutine reads and interpolates these profiles for photometric calculations. Magnitudes are computed in the AB system and output spectra are provided in units of solar luminosity per hertz ($L_\odot$/Hz).

10. Spectral Indices

The SPS model computes spectral absorption-line indices from the synthetic spectra. These indices are widely used to constrain the age, metallicity, and chemical composition of stellar populations and galaxies.

The index definitions are provided in the file indices.dat, in the INDICES/ directory.

The implemented indices include several commonly used systems such as:

- Lick indices
- DTT indices
- BH indices
- CaHK and CaII triplet indices
- Conroy & van Dokkum (CvD) indices
- SDSS-based NaI and TiO indices

The computation of indices is controlled through the parameter index_cal:

= 0 : Do not compute spectral indices.
= 1 : Compute spectral indices.

11. OUTPUTS:

The SPS-VARIMF code produces two main output files:

`*.mags` : Integrated physical and photometric properties as a function of time.
`*.spec` : Spectral energy distributions (SEDs) as a function of time.

The output filename begins with **SSP** or **CSP**, indicating whether a Simple Stellar Population or Composite Stellar Population model was computed. The filename also includes the adopted IMF and, for CSP models, the selected star formation history (SFH).

Like “CSP_IGIMF_ConstSFR_.mags”

Unless otherwise specified, all masses, luminosities, remnant numbers, and related quantities are normalized such that the total stellar mass formed over the entire lifetime of the model galaxy is equal to one solar mass.

M_{tot}=1 M⊙

To obtain physical quantities for a galaxy with a different total formed stellar mass, all extensive quantities (e.g., masses, luminosities, remnant numbers) should be scaled accordingly.

The *.mags File

The `*.mags` file contains the time evolution of the integrated physical, chemical, and photometric properties of the model.

Each row corresponds to one evolutionary time step and contains:

Column	Description
log(age)	Logarithm of the population age (yr).
log(mass)	Logarithm of the mass locked in living stars and stellar remnants (M⊙).
log(Lbol)	Logarithm of the bolometric luminosity (L⊙).
log(SFR)	Logarithm of the star formation rate (M⊙/yr). For SSP models this quantity is not applicable after the initial burst.
log(mass_rem)	Logarithm of the total stellar remnant mass (M⊙).
log(mass_wd)	Logarithm of the total white dwarf mass (M⊙).
Z_gas	Mass-weighted metallicity of the gas component.
Z_star	Mass-weighted metallicity of the stellar component.
Mgas_csp	Remaining gas mass in the CSP model (M⊙).
num_SNIa	Cumulative number of Type Ia supernovae (10^{-4}).
num_NSs	Total number of neutron stars.
num_BHs	Total number of black holes.
mass_NSs	Total mass contained in neutron stars (M⊙).
mass_BHs	Total mass contained in black holes (M⊙).
Magnitudes	Integrated magnitudes in all filters specified in FILTER_LIST.

The *.spec File

The first line of the file contains the wavelength grid in Angstroms, covering the range from 91 to (10^8) (\AA).

The remainder of the file is organized into two-line blocks, one block for each evolutionary time step:

```
log(age)  log(mass)  log(Lbol)  log(SFR)
Fν(λ1)  Fν(λ2)  Fν(λ3)  ...  Fν(λN)
```

The first line contains the basic properties of the stellar population at the corresponding age, while the second line contains the spectral luminosity density (F_\nu) at each wavelength listed in the header.

The spectra are given in units of (L_sun/Hz). Thus, each block provides the complete SED of the stellar population at a given age.

The ELEMENTS.txt File

The code produces an `ELEMENTS.txt` file containing the chemical composition of the gas and stellar components as a function of time.

Each row corresponds to one evolutionary time step and contains:

```
log(age)
Z_gas  Ca_g  C_g  Fe_g  He_g  H_g  Mg_g  Ne_g  N_g  O_g  S_g  Si_g
Z_star Ca_s  C_s  Fe_s  He_s  H_s  Mg_s  Ne_s  N_s  O_s  S_s  Si_s
```

where `Z_gas` and `Z_star` are the total metallicities of the gas and stellar components, respectively. The subscripts `_g` and `_s` denote abundances in the gas and stellar components.

The elemental quantities are given as mass fractions, representing the mass of each element divided by

the total mass of the corresponding component. The file therefore follows the abundances of H, He, C, N, O, Ne, Mg, Si, S, Ca, and Fe in both the gas and stellar populations.

For models with `Z_MODE = 0`, the adopted metallicity remains fixed and no self-consistent chemical evolution is calculated. In this case, the abundances reflect the fixed input chemical composition. For models with `Z_MODE = 1`, the abundances evolve self-consistently according to the chemical enrichment history of the model.

The *.indx File

When `index_cal = 1`, SPS-VARIMF calculates the selected spectral indices from the model spectrum and writes the results to an `*.indx` output file. When `index_cal = 0`, spectral indices are not calculated and the `*.indx` file is not generated.

The spectral indices to be calculated are specified in:

`INDICES/list_indices`

This file contains the names of the available spectral indices and their corresponding reference numbers. The available indices include the classical **Lick indices**, together with several additional optical and near-infrared indices, such as **DTT**, **BH**, **Ca II**, **Paschen**, **Na I**, **Mg I**, **FeH**, **TiO**, and **CaH** indices.

The definitions and wavelength ranges used to calculate the indices are provided in:

`INDICES/indices.dat`

For each index, `indices.dat` specifies the central index passband, the blue and red continuum sidebands, the units of the index, and whether the wavelengths are given in air or vacuum.

The `*.indx` file contains the calculated values of all indices selected in `INDICES/list_indices` at each evolutionary time step.

The N_OBstars.txt File

When `OB_stars = 1`, SPS-VARIMF calculates the number of O- and B-type stars in several initial-mass ranges and writes the results to the `N_OBstars.txt` output file. When `OB_stars = 0`, this file is not generated.

The file provides the predicted numbers of O- and B-type stars divided into five initial-mass ranges:

Stellar type	Initial-mass ranges (M_sun)
O stars	15–30, 30–50, 50–70, 70–90, 90–150
B stars	2–5, 5–7, 7–9, 9–11, 11–15

The numbers are calculated by summing the model stellar numbers over the corresponding initial-mass bins. Thus, `N_OBstars.txt` provides a useful diagnostic of the massive-star content predicted by the adopted IMF and star-formation history.

The O-star population is particularly relevant for studies of ionizing-photon production and Hα emission, while the B-star population provides additional information on the intermediate-to-high-mass stellar component and the UV/optical stellar population.

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12. Future Enhancements

- **Gas Inflow**: Current metallicity evolution is limited to a closed-box model, meaning no gas inflow or outflow occurs.. Gas inflow will be added to account for external enrichment.
- **Additional SFHs**: Support for more complex SFHs, including bursty star formation.

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13. Acknowledgments

This program is built upon extensive research in stellar population synthesis and uses publicly available codes and libraries, including Padova, PARSEC, BaSeL, GALAXEV (Bruzual & Charlot), FSPS (Flexible Stellar Population Synthesis), and others. We acknowledge the invaluable contributions of the researchers and developers who have made these resources available to the scientific community.

SPS-VARIMF was developed by **Akram Hasani Zonoozi** at the *Institute for Advanced Studies in Basic Sciences (IASBS)*. For any publications or research that utilize **SPS-VARIMF** , please cite the following reference:

Zonoozi, Haghi, and Kroupa et al. MNRAS, 2025 (https://doi.org/10.1093/mnras/staf091)
"Stellar population synthesis models with a physically varying IMF"

For comments, questions, bug reports, or feedback, please contact:

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Table1: Metallicity Index (`zmet_ini`)

Z_index	Z [log(Z/Z⊙)]
1	0.0002 (-1.98)
2	0.0003 (-1.80)
3	0.0004 (-1.68)
4	0.0005 (-1.58)
5	0.0006 (-1.50)
6	0.0008 (-1.38)
7	0.0010 (-1.28)
8	0.0012 (-1.20)
9	0.0016 (-1.07)
10	0.0020 (-0.98)
11	0.0025 (-0.89)
12	0.0031 (-0.79)
13	0.0039 (-0.69)
14	0.0049 (-0.59)
15	0.0061 (-0.49)
16	0.0077 (-0.39)
17	0.0096 (-0.30)
18	0.0120 (-0.20)
19	0.0150 (-0.10)
20	,0.0190 (+0.00)
21	0.0240 (+0.10)
22	0.0300 (+0.20)